GLOBAL EXISTENCE OF SMOOTH SELF-SIMILAR QUENCHING PROFILES FOR $u_{tt} = \Delta u - u^{-2}$: A RIGOROUS VERSION OF THEOREM 2.1 OF GIMPERLEIN–HE–LACEY

RUNAN HE

Abstract. In [7] the existence of a sequence of smooth, radial, global self-similar quenching profiles for the semilinear wave equation $u_{tt} = \Delta u - u^{-2}$ in dimensions $2 \le n \le 7$ was obtained formally (Theorem 2.1 there), by matched asymptotics applied to the profile equation

$$(1-\eta^2)v'' + \Big(\frac{n-1}{\eta} - \frac{2}{3}\eta\Big)v' + \frac{2}{9}v - \frac{1}{v^2} = 0, \qquad v(0) = c, \quad v'(0) = 0.$$

This note gives a complete proof. The argument combines three ingredients that are by now standard in the blow-up literature following Merle and his coauthors: (i) the light cone $\eta = 1$ is treated as a *sonic point* in the sense of Merle–Raphaël–Rodnianski–Szeftel, where a non-resonance condition ($\alpha = \frac{n}{2} + \frac{1}{6} \notin \mathbb{N}$) singles out a unique analytic branch through the cone and defines a scalar *singular coefficient* $\beta(c)$; (ii) a quantitative inner/outer matching in the spirit of Collot–Raphaël–Szeftel (and of the earlier shooting constructions of Budd–Qi, Lepin and Bizoń–Maison–Wasserman), in which the exact singular profile $v_{\mathrm{s}}(\eta) = a\eta^{2/3}$ plays the role of the singular self-similar solution and the inner Emden–Fowler-type profile oscillates around it with a logarithmic phase; (iii) a weighted flux identity, the ordinary-differential analogue of the Merle–Zaag energy identity in similarity variables, which yields monotonicity, global continuation beyond the cone and the sharp growth $v(\eta) = b\,\eta^{2/3} + O(\eta^{-1/3})$ at infinity. The smooth global profiles are exactly the zeros of β, and $\beta(c) = Kc^{\frac{3n}{4}-\frac{1}{2}}\big[\sin\big(\frac{3b}{2}\ln\frac{1}{c} - \vartheta\big) + o(1)\big]$ as $c \to 0$ with $K \neq 0$, which produces the discrete sequence $c_k \to 0$, $\ln c_k = -\frac{2\pi}{3b}k + \mathrm{const} + o(1)$. Numerical two-point shooting confirms the picture for $n = 2, \dots, 7$; in particular consecutive profiles are separated by *one* additional intersection with v_{s}, so the spacing of $\ln c_k$ is $2\pi/(3b)$ rather than the $4\pi/(3b)$ suggested by (2.33) of [7].

Contents

1. Setting and main result 2
1.1. The profile equation 2
1.2. Logarithmic variables and the weighted flux identity 2
1.3. Smooth solutions and the main theorem 3
2. The light cone as a sonic point 4
2.1. Every solution reaches the cone 4
2.2. Flux variables and the Briot–Bouquet structure 4
2.3. The singular coefficient 5
2.4. The linearised equation at the singular profile 6
3. Beyond the light cone: monotonicity, global continuation and growth 7
4. Inner region and matching 8
4.1. The inner profile 8
4.2. The inner region of v_c 9
4.3. The singular coefficient near the singular profile 9
4.4. Matching: asymptotics of $\beta(c)$ 10
5. Proof of Theorem 1.2 11

Date: October 2026.

6. Numerical confirmation 12
Acknowledgement 12
References 12

1. Setting and main result

1.1. **The profile equation.** Radial self-similar solutions $u(x,t) = (T-t)^{2/3}v(\eta)$, $\eta = |x|/(T-t)$, of $u_{tt} = \Delta u - u^{-2}$ in $\mathbb{R}^n$ are governed by [7, (2.3)–(2.4)]

$$(1-\eta^2)v'' + \Big(\frac{n-1}{\eta} - \frac{2}{3}\eta\Big)v' + \frac{2}{9}v - \frac{1}{v^2} = 0, \qquad \eta > 0, \tag{1}$$

$$v(0) = c > 0, \qquad v'(0) = 0. \tag{2}$$

Throughout, $n \geq 2$ and we use the constants

$$c^* = \Big(\frac{9}{2}\Big)^{1/3}, \qquad a = \Big[\frac{2}{3}\Big(n - \frac{4}{3}\Big)\Big]^{-1/3}, \qquad \alpha = \frac{n}{2} + \frac{1}{6},$$
$$\nu = \frac{n}{2} - \frac{1}{3}, \qquad b = \sqrt{\frac{8}{3} - \Big(\frac{n}{2} - \frac{7}{3}\Big)^2} \quad (2 \leq n \leq 7). \tag{3}$$

Two explicit solutions of (1) will be used constantly.

Lemma 1.1 (two exact solutions)**.** *$v \equiv c^*$ solves (1)–(2) with $c = c^*$. The* singular profile

$$v_{\mathrm{s}}(\eta) = a\,\eta^{2/3} \tag{4}$$

solves (1) on all of $(0,\infty)$, through the light cone, but is not regular at $\eta = 0$. Moreover $v_{\mathrm{s}} < c^*$ exactly on $(0,\eta^*)$ with $\eta^* = (c^*/a)^{3/2} = (3n-4)^{1/2} > 1$.

Proof. For v_{s}: $(1-\eta^2)v_{\mathrm{s}}'' + (\frac{n-1}{\eta} - \frac{2}{3}\eta)v_{\mathrm{s}}' + \frac{2}{9}v_{\mathrm{s}} = a\eta^{-4/3}\big[-\frac{2}{9} + \frac{2}{9}\eta^2 + \frac{2}{3}(n-1) - \frac{4}{9}\eta^2 + \frac{2}{9}\eta^2\big] = \frac{2}{3}\big(n - \frac{4}{3}\big)a\,\eta^{-4/3}$, and $v_{\mathrm{s}}^{-2} = a^{-2}\eta^{-4/3}$; the two agree precisely when $a^3 = [\frac{2}{3}(n-\frac{4}{3})]^{-1}$. The η^2-terms cancel, which is why v_{s} passes through $\eta = 1$. Finally $(c^*/a)^3 = \frac{9}{2}\cdot\frac{2}{3}(n-\frac{4}{3}) = 3n-4$. □

1.2. **Logarithmic variables and the weighted flux identity.** Set $\eta = e^s$ and $Y(s) = \eta^{-2/3}v(\eta)$, i.e. $v = \eta^{2/3}Y$. A direct computation (the e^{2s}-terms in the zeroth-order coefficient cancel, as in Lemma 1.1) turns (1) into

$$(1 - e^{2s})\,Y_{ss} + \Big(n - \frac{2}{3} - e^{2s}\Big)Y_s + G'(Y) = 0, \qquad G(Y) = \frac{1}{3}\Big(n - \frac{4}{3}\Big)Y^2 + \frac{1}{Y}, \tag{5}$$

so that $G'(Y) = \frac{2}{3}(n - \frac{4}{3})Y - Y^{-2}$ vanishes exactly at $Y = a$, and $G''(a) = 2(n - \frac{4}{3}) > 0$. The constant $Y \equiv a$ is v_{s}; the constant solution c^* is $Y = c^*e^{-2s/3}$. As $s \to -\infty$ the non-autonomous terms disappear and (5) reduces to the autonomous *inner equation*

$$Y_{ss} + \Big(n - \frac{2}{3}\Big)Y_s + G'(Y) = 0, \tag{6}$$

which is [7, (2.14)]. The equation (6) has the Lyapunov function $L = \frac{1}{2}Y_s^2 + G(Y)$, $\frac{d}{ds}L = -(n - \frac{2}{3})Y_s^2$, and the linearisation at the equilibrium $(a,0)$ has eigenvalues

$$\lambda_\pm = -\nu \pm \sqrt{\Big(\frac{n}{2} - \frac{7}{3}\Big)^2 - \frac{8}{3}} = -\nu \pm ib \qquad (2 \leq n \leq 7), \tag{7}$$

so $(a,0)$ is a stable focus precisely for $2 \leq n \leq 7$, cf. [7, (2.17)].

The second structural fact is a weighted flux identity, which is the one-dimensional analogue of the Merle–Zaag energy identity in similarity variables [14, 15]: with

$$\rho(\eta) = \eta^{n-1}|1-\eta^2|^{-\alpha}, \qquad \Phi(\eta) = \rho(\eta)(1-\eta^2)v'(\eta) = \eta^{n-1}\,\mathrm{sgn}(1-\eta^2)|1-\eta^2|^{1-\alpha}v'(\eta), \tag{8}$$

equation (1) is equivalent, on each side of the cone, to

$$\Phi'(\eta) = \rho(\eta)\Big(\frac{1}{v^2} - \frac{2}{9}v\Big). \tag{9}$$

(Indeed $\rho'/\rho = \frac{n-1}{\eta} + \frac{(n+\frac{1}{3})\eta}{1-\eta^2}$ is exactly what is needed for $(\rho(1-\eta^2)v')' = \rho[(1-\eta^2)v'' + (\frac{n-1}{\eta} - \frac{2}{3}\eta)v']$.) The weight ρ is the formal Merle–Zaag weight $(1-|y|^2)^{\frac{2}{p-1}-1}$ for the "exponent" $p=-2$ of the nonlinearity $-u^{-2}$; it is not integrable at the cone (since $\alpha > 1$), which is the reason why no Lyapunov functional of Merle–Zaag type exists for (5), as observed in [7]. Nevertheless (9) encodes the sonic-point structure (Section 2) and the monotonicity properties (Section 3).

1.3. **Smooth solutions and the main theorem.** By a *solution* of (1)–(2) on $[0,\eta_1)$ we mean a positive C^2 function satisfying (1) pointwise for $0<\eta<\eta_1$, $\eta\neq 1$, and (2); at $\eta=1$ the second-order term drops out and (1) reduces to $v'(1) = p(v(1))$, where

$$p(c_1) := \frac{c_1^{-2} - \frac{2}{9}c_1}{n - \frac{5}{3}}. \tag{10}$$

By Proposition A.1 of [7] (local existence at $\eta=0$ and at $\eta=1$) every $c>0$ determines a unique solution v_c on a maximal interval $[0,\eta_{\max}(c))$, $\eta_{\max}\le 1$ possibly, and $v_c \in C^\infty([0,\eta_{\max}))$ if $\eta_{\max}\le 1$. We call v a *smooth global solution* if it is a solution on $[0,\infty)$ and $v\in C^\infty([0,\infty))$.

Theorem 1.2 (rigorous form of [7, Theorem 2.1])**.** *Let $2\le n\le 7$. There exist $c_0\in(0,c^*)$ and an index $k_0\in\mathbb{N}$ with the following properties.*

(i) *The set $\mathcal{C} = \{c\in(0,c_0] :\ v_c$ is a smooth global solution$\}$ is discrete in $(0,c_0]$ and accumulates at 0: there are a phase $\vartheta\in\mathbb{R}$ (depending only on n) and, for every $k\ge k_0$, a nonempty window*

$$J_k = \Big\{c>0 :\ \Big|\tfrac{3b}{2}\ln\tfrac{1}{c} - \vartheta - \pi k\Big| \le \varpi_k\Big\}, \qquad \varpi_k\downarrow 0,$$

such that $\mathcal{C}\cap(0,c_0]\subset\bigcup_{k\ge k_0} J_k$ and $\mathcal{C}\cap J_k\neq\emptyset$ for every $k\ge k_0$. In particular, choosing $c_k\in\mathcal{C}\cap J_k$,

$$\ln c_k = -\frac{2\pi}{3b}\,k + \frac{2\vartheta}{3b} + o(1) \quad (k\to\infty), \qquad \text{so} \qquad \frac{c_{k+1}}{c_k}\to e^{-2\pi/(3b)}. \tag{11}$$

(ii) *Each $v_k := v_{c_k}$ is real-analytic on $(0,\infty)$ (in particular across the light cone), strictly increasing, and $v_k<c^*$ on $[0,\eta_k^{**})$, $v_k>c^*$ on (η_k^{**},∞) with $\eta_k^{**}\to\eta^* = (3n-4)^{1/2} > 1$.*

(iii) *There are constants $b_k>0$, $b_k\to a$, with $v_k(\eta) = b_k\eta^{2/3} + O(\eta^{-1/3})$ as $\eta\to\infty$; more precisely $\eta^{-2/3}v_k(\eta)$ is a real-analytic function of $1/\eta$ near $1/\eta=0$.*

(iv) *(Inner and intermediate behaviour.) Let V_0 be the solution of $V''+\frac{n-1}{r}V' = V^{-2}$, $V(0)=1$, $V'(0)=0$. Then V_0 is global, increasing, and $V_0(r) = ar^{2/3} + A_1 r^{1-n/2}\cos(b\ln r + A_2) + O(r^{1-n/2-\nu})$ as $r\to\infty$ with $A_1>0$; and for every $\varepsilon>0$ there are $S_0, c_0'>0$ such that for $c\in\mathcal{C}\cap(0,c_0']$*

$$\Big|\frac{v_c(\eta)}{c\,V_0(\eta/c^{3/2})} - 1\Big| \le \varepsilon \quad \text{for } 0\le\eta\le c^{3/2}e^{S_0}, \qquad \Big|\frac{v_c(\eta)}{a\eta^{2/3}} - 1\Big| \le \varepsilon \quad \text{for } c^{3/2}e^{S_0}\le\eta\le 1+\delta_0.$$

In particular $v_k(\eta)\sim a\eta^{2/3}$ in the intermediate region $c_k^{3/2}\ll\eta\ll 1$, which is part (b) of [7, Theorem 2.1].

(v) *If $c\in J_k\cap\mathcal{C}$ and $c'\in J_{k+1}\cap\mathcal{C}$, then $v_{c'}-v_{\mathrm{s}}$ has exactly one more zero in $(0,1)$ than $v_c - v_{\mathrm{s}}$; thus k labels the number of intersections of v_k with the singular profile, up to a fixed shift.*

Remark 1.3. (a) The phase ϑ is explicit in terms of two universal quantities: the phase A_2 of the inner profile V_0 and the phase of the solution of the linearised equation at v_{s} which is regular at the cone (Proposition 2.7 below): $\vartheta = B_2 - A_2$, see (21). This is the content of the formal matching (2.31)–(2.33) in [7]; the only difference is that the matching condition holds modulo π and not modulo 2π, because the regular linear mode is determined only up to a real factor, so that consecutive profiles are separated by one (not two) additional intersection with v_{s} (Table 1).

(b) With a C^1 version (in $\ln c$) of the matching estimate of Proposition 4.4 one obtains that β has exactly one, transversal, zero in each window J_k, so that $\mathcal{C} \cap (0, c_0] = \{c_k\}_{k \geq k_0}$. This requires differentiating the whole construction with respect to the parameter and is not needed for Theorem 1.2; we comment on it in Remark 4.5.

2. The light cone as a sonic point

The guiding principle of this section is borrowed from Merle–Raphaël–Rodnianski–Szeftel [12, 13]: a degenerate point of the profile equation (there the sonic point of the self-similar Euler flow, here the light cone $\eta = 1$) is an equilibrium of a desingularised dynamical system, and smoothness of the profile across it is a codimension-one condition which is non-degenerate exactly when the ratio of the two relevant eigenvalues is not an integer. Here the eigenvalues are 2 and $n - \frac{5}{3}$, and the ratio $\alpha - 1 = \frac{n}{2} - \frac{5}{6}$ is never an integer.

2.1. Every solution reaches the cone.

Lemma 2.1 (energy inside the cone)**.** *For a solution of* (1) *on an interval of* $(0,1)$ *the quantity* $E(\eta) = \frac{1}{2}(1-\eta^2)v'(\eta)^2 + \frac{1}{v(\eta)} + \frac{v(\eta)^2}{9}$ *satisfies* $E'(\eta) = -\big(\frac{n-1}{\eta} + \frac{\eta}{3}\big)v'(\eta)^2 \leq 0$. *Consequently, for every* $c > 0$ *the solution* v_c *of* (1)–(2) *exists on* $[0,1)$ *and, with* $E_0 = c^{-1} + c^2/9$,

$$E_0^{-1} \leq v_c(\eta) \leq 3E_0^{1/2}, \qquad |v_c'(\eta)| \leq (2E_0)^{1/2}(1-\eta^2)^{-1/2} \qquad (0 \leq \eta < 1). \tag{12}$$

In particular $v_c(1) := \lim_{\eta \to 1^-} v_c(\eta)$ *exists and is positive.*

Proof. Using $(1-\eta^2)v'' = -(\frac{n-1}{\eta} - \frac{2}{3}\eta)v' + q(v)$ with $q(v) = v^{-2} - \frac{2}{9}v = \frac{d}{dv}(-\frac{1}{v} - \frac{v^2}{9})$ one finds $E' = -\eta v'^2 + v'(1-\eta^2)v'' - q(v)v' = -(\eta + \frac{n-1}{\eta} - \frac{2}{3}\eta)v'^2$. The bounds follow from $E \leq E(0) = E_0$, and v' is integrable on $[0,1)$ by the last bound. □

2.2. Flux variables and the Briot–Bouquet structure.

Put $x = 1 - \eta \in (0, \delta)$ and, with Φ as in (8),

$$\psi(x) := x^{\alpha-1}\Phi(1-x), \qquad h(x) := (1-x)^{1-n}(2-x)^{\alpha-1}, \qquad g(x) := (1-x)^{n-1}(2-x)^{-\alpha}. \tag{13}$$

Then (9) is equivalent to the first-order system

$$x\frac{dv}{dx} = -x\,h(x)\,\psi, \qquad x\frac{d\psi}{dx} = (\alpha-1)\psi - g(x)\,q(v), \qquad q(v) = v^{-2} - \tfrac{2}{9}v. \tag{14}$$

This is a Briot–Bouquet system $x\,\mathbf{y}' = \mathbf{F}(x,\mathbf{y})$, $\mathbf{y} = (v,\psi)$, with $\mathbf{F}$ analytic in a neighbourhood of $\{x = 0,\ v > 0\}$. Its equilibria at $x = 0$ form the curve $\psi = \psi_1(v) := 2^{-\alpha}q(v)/(\alpha - 1)$, and

$$D_{\mathbf{y}}\mathbf{F}(0, v, \psi_1(v)) = \begin{pmatrix} 0 & 0 \\ -2^{-\alpha}q'(v) & \alpha - 1 \end{pmatrix}, \qquad \text{eigenvalues } 0 \ \text{ and } \ \alpha - 1 = \frac{n}{2} - \frac{5}{6} \notin \mathbb{N}.$$

Note that $v'(1) = h(0)\psi_1(v(1)) = p(v(1))$, see (10).

Lemma 2.2 (arrival at the cone in flux variables)**.** *Let* v *be a solution of* (1) *on* $[1-\delta, 1)$ *with* $\inf v > 0$. *Then* $c_1 = v(1)$ *exists,* $\psi(x) \to \psi_1(c_1)$ *and* $v'(\eta) \to p(c_1)$ *as* $x = 1 - \eta \to 0$.

Proof. Lemma 2.1 gives the existence of $c_1 > 0$. From (14), $(x^{1-\alpha}\psi)' = -x^{-\alpha}g(x)q(v(x))$, hence $x^{1-\alpha}\psi(x) = x_0^{1-\alpha}\psi(x_0) + \int_x^{x_0} \zeta^{-\alpha}g(\zeta)q(v(\zeta))\,\mathrm{d}\zeta$. Since $\alpha > 1$ and $g(\zeta)q(v(\zeta)) \to 2^{-\alpha}q(c_1)$, the integral equals $2^{-\alpha}q(c_1)x^{1-\alpha}/(\alpha-1) + o(x^{1-\alpha})$, which gives $\psi \to \psi_1(c_1)$; then $v' = -v_x = h\psi \to h(0)\psi_1(c_1) = p(c_1)$. □

Lemma 2.3 (the analytic branch through the cone)**.** *For every $c_1 > 0$ there is exactly one solution $v_{\mathrm{reg}}(\cdot; c_1)$ of (1) which is real-analytic in a neighbourhood of $\eta = 1$ and satisfies $v_{\mathrm{reg}}(1; c_1) = c_1$. It is real-analytic jointly in (η, c_1), and $v_{\mathrm{reg}}(\cdot; a) = v_{\mathrm{s}}$.*

Proof. Since no eigenvalue of $D_{\mathbf{y}}\mathbf{F}(0, c_1, \psi_1(c_1))$ is a positive integer, the Briot–Bouquet theorem [8, Ch. 11] (see also [9, Ch. 2]) provides a unique solution of (14) analytic at $x = 0$ with $\mathbf{y}(0) = (c_1, \psi_1(c_1))$; appending the equation $x\,dc_1/dx = 0$ shows analytic dependence on c_1. The same holds on the right of the cone (replace x by $\eta - 1$), and both expansions are generated by the same recursion, so they define one analytic function near $\eta = 1$. Concretely, writing $v = \sum_m t_m(\eta-1)^m$ and inserting into (1), the coefficient of $(\eta-1)^m$ determines t_{m+1} through the factor $(m+1)\big(n - \frac{5}{3} - 2m\big) = 2(m+1)(\alpha - 1 - m)$, which never vanishes; this is the non-resonance condition. Uniqueness of the analytic solution shows $v_{\mathrm{reg}}(\cdot; a) = v_{\mathrm{s}}$. □

2.3. **The singular coefficient.**

Proposition 2.4 (structure of all solutions at the cone)**.** *Let v be a solution of (1) on $[1-\delta, 1)$ with $\inf v > 0$ and $c_1 = v(1)$. There is a unique $\beta \in \mathbb{R}$ such that, as $\eta \to 1^-$,*

(15)
$$v(\eta) - v_{\mathrm{reg}}(\eta; c_1) = \beta\,(1-\eta)^{\alpha}\big(1 + o(1)\big), \qquad v'(\eta) - v'_{\mathrm{reg}}(\eta; c_1) = -\alpha\beta\,(1-\eta)^{\alpha-1}\big(1 + o(1)\big).$$

Moreover v extends to a function of class $C^{\lfloor\alpha\rfloor+1}$ on $[1-\delta, 1]$ if and only if $\beta = 0$, and then $v = v_{\mathrm{reg}}(\cdot; c_1)$ on $[1-\delta, 1)$; in particular v extends to a C^∞ (indeed analytic) solution across the cone if and only if $\beta = 0$.

Proof. Let $u = v - v_{\mathrm{reg}}$, $\psi_u = \psi - \psi_{\mathrm{reg}}$, with ψ_{reg} the flux variable of v_{reg}. By Lemma 2.2, $u(0) = 0$ and $\psi_u(0) = 0$ (as functions of x), and from (14)

(16)
$$u(x) = -\int_0^x h\,\psi_u\,\mathrm{d}\zeta, \qquad (x^{1-\alpha}\psi_u)' = -x^{-\alpha}g(x)Q(x)u(x), \qquad Q = \frac{q(v) - q(v_{\mathrm{reg}})}{v - v_{\mathrm{reg}}} \in C([0,\delta]).$$

Let $m(x) = \sup_{[0,x]}|\psi_u|$, so that $|u(\zeta)| \le C\zeta\,m(\zeta)$. Integrating the second relation from x to x_0,

$$\psi_u(x) = x^{\alpha-1}\Big[x_0^{1-\alpha}\psi_u(x_0) + \int_x^{x_0} \zeta^{-\alpha}gQu\,\mathrm{d}\zeta\Big].$$

If $\alpha < 2$ the integral converges at 0 and $\psi_u(x) = \beta' x^{\alpha-1} + O(x)$, with $\beta' = x_0^{1-\alpha}\psi_u(x_0) + \int_0^{x_0}\zeta^{-\alpha}gQu$. If $\alpha > 2$, the bound $|u| \le C\zeta m(x_0)$ gives $\psi_u = O(x)$, hence $|u| \le Cx^2$, hence $\psi_u = \beta' x^{\alpha-1} + O(x^2)$ if $\alpha < 3$, and so on; after $\lfloor\alpha\rfloor$ steps one arrives at

$$\psi_u(x) = \beta' x^{\alpha-1} + O(x^{\lfloor\alpha\rfloor}), \qquad \text{whence} \qquad u(x) = -\frac{2^{\alpha-1}\beta'}{\alpha}x^{\alpha} + O(x^{\lfloor\alpha\rfloor+1}), \tag{17}$$

and (15) holds with $\beta = -2^{\alpha-1}\beta'/\alpha$ (recall $\alpha \notin \mathbb{N}$, so x^α and $x^{\lfloor\alpha\rfloor+1}$ are distinguishable). If $\beta' = 0$ the bootstrap continues indefinitely, $\psi_u = O(x^N)$ for all N, and then the weighted estimate $\sup_{[0,x]}\zeta^{-N}|\psi_u| \le \frac{C}{N-\alpha+1}\sup_{[0,x]}\zeta^{-N}|u| \le \frac{C'x}{N-\alpha+1}\sup_{[0,x]}\zeta^{-N}|\psi_u|$ forces $\psi_u \equiv 0$, i.e. $u \equiv 0$. Finally, if $v \in C^{\lfloor\alpha\rfloor+1}([1-\delta,1])$ then u has a Taylor polynomial P of degree $\lfloor\alpha\rfloor$ at $x = 0$ with $u = P + O(x^{\lfloor\alpha\rfloor+1})$; comparing with $u = \beta x^\alpha(1 + o(1))$ and using $\alpha \notin \mathbb{N}$ gives first $P \equiv 0$ and then $\beta = 0$. □

Definition 2.5 (shooting functional)**.** For $c > 0$ let $\beta(c)$ be the number β of Proposition 2.4 for $v = v_c$ (which reaches the cone by Lemma 2.1). Thus v_c extends to a C^∞ solution across the light cone if and only if $\beta(c) = 0$, and then the extension is $v_{\mathrm{reg}}(\cdot; v_c(1))$.

Lemma 2.6. *β is real-analytic on $(0, \infty)$, and $\beta(c^*) = 0$.*

Proof. By the Briot–Bouquet theory for a positive non-integer eigenvalue [8, Ch. 12], the solutions of (14) converging to $(c_1, \psi_1(c_1))$ form a one-parameter family

$$\mathbf{y}(x; c_1, \beta') = \sum_{j,k\ge 0} \mathbf{a}_{jk}(c_1, \beta')\, x^{j+k(\alpha-1)},$$

convergent for $0 < x < x_1$ uniformly for (c_1, β') in compacts, with $\mathbf{a}_{jk}$ analytic and $\mathbf{a}_{01} = (0, \beta')$; by Proposition 2.4 these are all solutions with $\inf v > 0$. The map $(c_1, \beta') \mapsto (v, \psi)(x_0)$ is analytic with invertible differential (its columns are the values at x_0 of the regular and the singular solution of the linearised system, which are independent), so by the inverse function theorem (c_1, β') is an analytic function of $(v_c(1-x_0), \psi_c(x_0))$, which in turn is analytic in c by analytic dependence of (1) on initial data on the regular interval $[0, 1-x_0]$ (the point $\eta = 0$ is handled as in [7, Lemma A.3]). Finally $v_{c^*} \equiv c^*$ is analytic, so $\beta(c^*) = 0$. □

2.4. **The linearised equation at the singular profile.** Write $Y = a + Z$ in (5). Then

(18)
$$\mathcal{L}Z := (1-e^{2s})Z_{ss} + \Big(n - \frac{2}{3} - e^{2s}\Big)Z_s + 2\Big(n - \frac{4}{3}\Big)Z = -N(Z), \qquad N(Z) = G'(a+Z) - G''(a)Z,$$

with $|N(Z)| + |N'(Z)|\,|Z| \le C_N |Z|^2$ for $|Z| \le a/2$. The linear operator $\mathcal{L}$ is [7, (2.29)–(2.30)] written for $Z = \eta^{-2/3} v_l$; note that the non-autonomous terms of $\mathcal{L}$ only involve Z_s and Z_{ss}, which is again the cancellation of Lemma 1.1.

Proposition 2.7 (a basis of $\ker \mathcal{L}$ on $(-\infty, 0)$)**.** *Let $2 \le n \le 7$.*

(a) *There are solutions Ψ_1, Ψ_2 of $\mathcal{L}Z = 0$ on $(-\infty, 0)$ with*

(19)
$$\begin{aligned} \begin{pmatrix} \Psi_1 \\ \Psi_1' \end{pmatrix} &= e^{-\nu s}\Big[\begin{pmatrix} \cos bs \\ -\nu \cos bs - b \sin bs \end{pmatrix} + O(e^{2s}) \Big], \\ \begin{pmatrix} \Psi_2 \\ \Psi_2' \end{pmatrix} &= e^{-\nu s}\Big[\begin{pmatrix} \sin bs \\ -\nu \sin bs + b \cos bs \end{pmatrix} + O(e^{2s}) \Big] \end{aligned}$$

as $s \to -\infty$.

(b) *$s = 0$ is a regular singular point of $\mathcal{L}$ with exponents 0 and α: every solution on $(-\delta, 0)$ is $A\,\chi_{\mathrm{reg}}(s) + B\,\chi_{\mathrm{sing}}(s)$ with χ_{reg} analytic at 0, $\chi_{\mathrm{reg}}(0) = 1$, and $\chi_{\mathrm{sing}}(s) = |s|^\alpha(1 + O(s))$. Let σ_j be the coefficient B of Ψ_j. Then $(\sigma_1, \sigma_2) \ne (0, 0)$.*

(c) *Write $(\sigma_1, \sigma_2) = \rho_0(\cos\varphi, \sin\varphi)$ with $\rho_0 > 0$. Then $\Psi_{\mathrm{reg}} := \sigma_2\Psi_1 - \sigma_1\Psi_2$ spans the solutions of $\mathcal{L}Z = 0$ on $(-\infty, 0]$ which are analytic at $s = 0$, and*

(20)
$$\Psi_{\mathrm{reg}}(s) = \rho_0 e^{-\nu s}\big[\cos(bs + B_2) + O(e^{2s})\big] \quad (s \to -\infty), \qquad B_2 := \frac{\pi}{2} - \varphi.$$

Proof. (a) For $s \le -1$ write $\mathcal{L}Z = 0$ as a first-order system $\mathbf{Z}' = (\mathbf{A} + \mathbf{R}(s))\mathbf{Z}$, $\mathbf{Z} = (Z, Z')$, where $\mathbf{A}$ is the constant matrix of the autonomous equation $Z'' + (n - \frac{2}{3})Z' + 2(n - \frac{4}{3})Z = 0$, with simple eigenvalues $\lambda_\pm = -\nu \pm ib$, and $\|\mathbf{R}(s)\| \le Ce^{2s}$. Since $\int_{-\infty}^{-1} \|\mathbf{R}\| < \infty$, Levinson's theorem [6, Thm. 1.3.1] (or [3, Ch. 3, Thm. 8.1]) yields solutions $\mathbf{Z}_\pm = e^{\lambda_\pm s}(\mathbf{e}_\pm + o(1))$, $\mathbf{e}_\pm$ the eigenvectors; the error is $O(\int_{-\infty}^{s} \|\mathbf{R}\|) = O(e^{2s})$ by the Volterra-equation form of the proof. Taking real and imaginary parts gives (19).

(b) Near $s = 0$ the coefficients of $\mathcal{L}$ are analytic, $1 - e^{2s} = -2s(1 + O(s))$, and the indicial equation of $\mathcal{L}$ at $s = 0$ is $2\lambda(\lambda - 1) - (n - \frac{5}{3})\lambda = 0$, i.e. $\lambda \in \{0, \alpha\}$; as $\alpha \notin \mathbb{Z}$ the Frobenius method gives the stated basis. If $\sigma_1 = \sigma_2 = 0$ then every solution of $\mathcal{L}Z = 0$ would be analytic at 0, contradicting the existence of χ_{sing}.

(c) is immediate from (a), (b): $\sigma_2 \cos bs - \sigma_1 \sin bs = \rho_0 \sin(\varphi - bs) = \rho_0 \cos(bs + \frac{\pi}{2} - \varphi)$. □

The number B_2 is the phase in [7, (2.31)] of the solution of the linearised equation "regular at $s = 0$". We can now define the phase of Theorem 1.2:

(21)
$$\vartheta := B_2 - A_2 \pmod{\pi},$$

where A_2 is the inner phase of Proposition 4.1 below.

3. Beyond the light cone: monotonicity, global continuation and growth

This section makes rigorous the second half of the formal proof in [7, §2(b)] and Lemma 2.4 there. No smallness is used; the input is only a sign configuration at the cone.

Lemma 3.1 (critical points, [7, §2(b)(a)–(d)]). *Let v solve (1) near $\eta_0 \neq 1$ with $v'(\eta_0) = 0$. Then $(1-\eta_0^2)v''(\eta_0) = q(v(\eta_0))$, $q(v) = v^{-2} - \frac{2}{9}v$. Hence, if $v(\eta_0) \neq c^*$, η_0 is a strict local extremum of v, namely*

	$0 < \eta_0 < 1$	$\eta_0 > 1$
$v(\eta_0) < c^*$	*minimum*	*maximum*
$v(\eta_0) > c^*$	*maximum*	*minimum*

and if $v(\eta_0) = c^*$ then $v \equiv c^*$.

Proof. Evaluate (1) at η_0; $q > 0$ below c^* and $q < 0$ above. The last claim is uniqueness for the regular initial value problem at $\eta_0 \neq 1$. □

Proposition 3.2 (global continuation beyond the cone). *Let v be a solution of (1) on $[1, \eta_1)$ which is C^1 at $\eta = 1$ with $v'(1) = p(v(1))$, and assume that $v(1) < c^*$ and that v attains the value c^* at some $\eta^{**} \in (1, \eta_1)$, with $v < c^*$ on $[1, \eta^{**})$. Then:*

(i) *$v' > 0$ on $[1, \eta^{**}]$;*
(ii) *v extends to a solution on $[1, \infty)$, with $v' > 0$ and $v > c^*$ on (η^{**}, ∞);*
(iii) *$Y(\eta) := \eta^{-2/3} v(\eta)$ converges to some $b \in (0, \infty)$, and Y is a real-analytic function of $t = 1/\eta$ in a neighbourhood of $t = 0$. In particular*

$$v(\eta) = b\,\eta^{2/3} + b_1 \eta^{-1/3} + O(\eta^{-4/3}) \qquad (\eta \to \infty), \tag{22}$$

and more generally v has a convergent expansion $\eta^{2/3} \sum_{j \geq 0} b_j \eta^{-j}$, which is [7, Lemma 2.4].

Proof. (i) $v'(1) = p(v(1)) > 0$ because $v(1) < c^*$. Let $\eta_2 = \sup\{\eta \in (1, \eta^{**}) : v' > 0 \text{ on } (1, \eta)\}$. If $\eta_2 < \eta^{**}$ then $v'(\eta_2) = 0$ with $v(\eta_2) < c^*$, so η_2 is a strict local maximum by Lemma 3.1; since $v(\eta^{**}) = c^* > v(\eta_2)$, v would then have a local minimum in (η_2, η^{**}) with value below c^*, which Lemma 3.1 forbids. Thus $v' > 0$ on $(1, \eta^{**})$, and $v'(\eta^{**}) = 0$ would force $v \equiv c^*$.

(ii) For $\eta > 1$ it is convenient to use the flux $\tilde{\Phi}(\eta) := \eta^{n-1}(\eta^2 - 1)^{1-\alpha} v'(\eta) = -\Phi(\eta)$, for which (9) reads

$$\tilde{\Phi}'(\eta) = \eta^{n-1}(\eta^2 - 1)^{-\alpha}\Big(\frac{2}{9}v - \frac{1}{v^2}\Big). \tag{23}$$

On any interval beyond η^{**} on which $v > c^*$ the flux $\tilde{\Phi}$ is increasing; since $\tilde{\Phi}(\eta^{**}) > 0$, v' cannot vanish there, so $v' > 0$ and $v > c^*$ persist as long as the solution exists. Blow-up of (v, v') in finite η is excluded because $v'' = [(\frac{n-1}{\eta} - \frac{2}{3}\eta)v' - q(v)]/(\eta^2 - 1)$ grows at most linearly in (v, v') on compact η-intervals away from $\eta = 1$, and $v > c^*$ keeps q bounded. Hence the solution is global.

(iii) *Step 1: $v \to \infty$.* Fix $\eta_3 > \eta^{**}$ and set $\kappa = \frac{2}{9}v(\eta_3) - v(\eta_3)^{-2} > 0$. By (23) and monotonicity of v, $\tilde{\Phi} \geq \tilde{\Phi}(\eta_3) > 0$ for $\eta \geq \eta_3$, so $v'(\eta) \geq \tilde{\Phi}(\eta_3)\eta^{1-n}(\eta^2 - 1)^{\alpha - 1} \geq c\,\eta^{-2/3}$ for large η (recall $2(\alpha - 1) = n - \frac{5}{3}$), and $v(\eta) \geq c'\eta^{1/3} \to \infty$.

Step 2: the equation for Y. Substituting $v = \eta^{2/3} Y$ in (1) (equivalently, writing (5) in the variable η) gives, for $\eta > 1$,

$$\eta^2(\eta^2 - 1)Y'' + \eta\Big(2\eta^2 - n - \frac{1}{3}\Big)Y' = G'(Y) = \frac{2}{3}\Big(n - \frac{4}{3}\Big)Y - \frac{1}{Y^2}, \tag{24}$$

and with the integrating factor $\mu(\eta) = \eta^{n + \frac{1}{3}}(\eta^2 - 1)^{1-\alpha} = \eta^2(1 + O(\eta^{-2}))$ this is

$$\big(\mu Y'\big)' = \frac{\mu(\eta)}{\eta^2(\eta^2 - 1)}\, G'(Y), \qquad \frac{\mu(\eta)}{\eta^2(\eta^2 - 1)} = \eta^{n - \frac{5}{3}}(\eta^2 - 1)^{-\alpha} = \eta^{-2}(1 + O(\eta^{-2})). \tag{25}$$

Since $G'(Y)$ has the sign of $Y-a$, a zero η_0 of Y' with $Y(\eta_0)>a$ (resp. $<a$) is followed by $Y'>0$ (resp. $Y'<0$) for all $\eta>\eta_0$, and $Y'(\eta_0)=0=Y(\eta_0)-a$ forces $Y\equiv a$. Hence Y is eventually strictly monotone and $b:=\lim Y\in[0,\infty]$ exists.

Step 3: $b<\infty$. If $Y\uparrow\infty$, then eventually $Y>a$, $Y'>0$, and by (25) $(\mu Y')'\le C\eta^{-2}Y$. Integrating and using monotonicity, $\mu Y'(\eta)\le C_1+C\,Y(\eta)/\eta_0$, so $Y'\le(C_1+C_2Y)\eta^{-2}$, which integrates to a bound on Y: contradiction.

Step 4: $b>0$. If $Y\downarrow 0$, then eventually $Y<a$, $\phi:=-\mu Y'>0$ is increasing by (25), and $Y(\eta)=\int_\eta^\infty \phi/\mu\ge\phi(\eta)\int_\eta^\infty\mu^{-1}\ge c\,\phi(\eta)/\eta$. Also $\phi'=\eta^{-2}(1+O(\eta^{-2}))\,(Y^{-2}-\frac{2}{3}(n-\frac{4}{3})Y)\le C\eta^{-2}Y^{-2}\le C'\phi^{-2}$, so $\phi^3\le C''\eta$ and therefore $Y(\eta)=\int_\eta^\infty\phi/\mu\le C\int_\eta^\infty\zeta^{1/3-2}\,\mathrm{d}\zeta=C\eta^{-2/3}$, i.e. $v=\eta^{2/3}Y$ is bounded, contradicting Step 1.

Step 5: asymptotics. With $b\in(0,\infty)$ the right-hand side of (25) is $O(\eta^{-2})$, so $\mu Y'\to m_1\in\mathbb{R}$, $Y'=m_1\eta^{-2}+O(\eta^{-3})$ and $Y=b-m_1\eta^{-1}+O(\eta^{-2})$, which is (22). Finally, in the variable $t=1/\eta$ equation (24) becomes

$$(1-t^2)\,Y_{tt}+\Big(n-\frac{5}{3}\Big)t\,Y_t=G'(Y),\tag{26}$$

which is a *regular* analytic equation at $t=0$ as long as $Y>0$. Since $(Y,Y_t)=(Y,-\eta^2Y')\to(b,-m_1)$ as $t\to0^+$, the solution extends analytically through $t=0$, which proves the last claim. (The exact solution $Y=c^*t^{2/3}$ shows that (26) is singular on $\{Y=0\}$; this is the alternative $v\to c^*$ in [7, Lemma 2.4], which is excluded for our solutions by Step 1.) □

Remark 3.3. Combining Lemma 2.1 with Proposition 3.2, the "possibility of limited existence as v could vanish at some finite value of η" noted in [7] can only occur outside the light cone, and it does not occur for solutions which cross c^* upwards after the cone.

4. Inner region and matching

We now make the matched asymptotics of [7, §2(a),(b)] quantitative, following the strategy of Collot–Raphaël–Szeftel [4] for the construction of self-similar profiles of the supercritical heat equation by bifurcation from the singular profile; the oscillatory situation at hand, in which the linearisation at the singular profile has complex eigenvalues, corresponds to the Sobolev-supercritical, Joseph–Lundgren-subcritical range of the heat equation, where the profiles were obtained by shooting by Troy, Budd–Qi and Lepin [16, 2, 11], and to the construction of self-similar wave profiles by Bizoń–Maison–Wasserman [1] and Dai–Duyckaerts [5]. Throughout, $2\le n\le 7$,

$$\sigma:=\tfrac{3}{2}\ln c\quad(\text{so } c^{3/2}=e^{\sigma}),\qquad Y_c(s):=e^{-2s/3}v_c(e^s),\qquad Z_c:=Y_c-a.$$

4.1. The inner profile.

Proposition 4.1 (inner profile)**.** *Let V_0 be the solution of $V''+\frac{n-1}{r}V'=V^{-2}$, $V(0)=1$, $V'(0)=0$, and $Y_0(s)=e^{-2s/3}V_0(e^s)$.*

(a) *V_0 exists on $[0,\infty)$, $V_0'>0$ on $(0,\infty)$, and $1\le V_0(r)\le 1+r^2/(2n)$.*
(b) *Y_0 solves the inner equation (6) and $(Y_0,Y_0')(s)\to(a,0)$ as $s\to\infty$.*
(c) *There are $A_1>0$ and $A_2\in\mathbb{R}$ such that, as $s\to\infty$,*

$$\begin{aligned} Y_0(s)&=a+A_1e^{-\nu s}\cos(bs+A_2)+O(e^{-2\nu s}),\\ Y_0'(s)&=A_1e^{-\nu s}\big[-\nu\cos(bs+A_2)-b\sin(bs+A_2)\big]+O(e^{-2\nu s}).\end{aligned}\tag{27}$$

Equivalently $V_0(r)=ar^{2/3}+A_1r^{1-n/2}\cos(b\ln r+A_2)+O(r^{1-n/2-\nu})$, which is [7, (2.21)].

Proof. (a) $(r^{n-1}V')'=r^{n-1}V^{-2}>0$ gives $V'>0$ and $V\ge1$, hence $(r^{n-1}V')'\le r^{n-1}$, $V'\le r/n$, and the solution is global.

(b) The computation leading to (5) with the e^{2s}-terms removed shows that Y_0 solves (6). Along solutions, $L=\frac12Y_s^2+G(Y)$ is nonincreasing, and $\{L\le L_0\}$ is a compact subset of

$\{Y>0\}\times\mathbb{R}$ because $G(Y)\to\infty$ as $Y\to 0^+$ or $Y\to\infty$. By LaSalle's invariance principle the ω-limit set of (Y_0,Y_0') is contained in the largest invariant subset of $\{Y_s=0\}$, which is the equilibrium $(a,0)$.

(c) Set $Z_0=Y_0-a$, so that $Z_0''+(n-\frac{2}{3})Z_0'+G''(a)Z_0=-N(Z_0)$ with N as in (18). The linear part is a hyperbolic sink with eigenvalues $-\nu\pm ib$, so (e.g. by a quadratic Lyapunov function for the linear part, perturbed by the $O(|Z_0|^3)$ cubic term once $|Z_0|$ is small) $|Z_0(s)|+|Z_0'(s)|\le C_\varepsilon e^{-(\nu-\varepsilon)s}$ for every $\varepsilon>0$. Let $K(s,s')=b^{-1}e^{-\nu(s-s')}\sin b(s-s')$ be the fundamental solution of the linear operator. For $s\ge s_0$,

$$Z_0(s)=e^{-\nu s}\big(C_1\cos bs+C_2\sin bs\big)-\int_{s_0}^{s}K(s,s')N(Z_0(s'))\,\mathrm{d}s',$$

and we split $\int_{s_0}^{s}=\int_{s_0}^{\infty}-\int_{s}^{\infty}$. Since $K(s,s')=e^{-\nu s}[\sin bs\cdot e^{\nu s'}\cos bs'-\cos bs\cdot e^{\nu s'}\sin bs']/b$ and $|N(Z_0(s'))|\le Ce^{-2(\nu-\varepsilon)s'}$, the first integral is $e^{-\nu s}$ times a convergent combination of $\cos bs,\sin bs$, which we absorb into (C_1,C_2), while the second is $O(e^{-\nu s}\int_s^\infty e^{\nu s'}e^{-2(\nu-\varepsilon)s'}\,\mathrm{d}s')=O(e^{-2(\nu-\varepsilon)s})$. This gives $|Z_0|\le Ce^{-\nu s}$, and running the argument once more, $Z_0=e^{-\nu s}(C_1\cos bs+C_2\sin bs)+O(e^{-2\nu s})$; the derivative is handled in the same way. It remains to see $(C_1,C_2)\neq 0$. Otherwise $Z_0=O(e^{-2\nu s})$, and iterating the previous step gives $Z_0=O(e^{-Ns})$ for every N, so that $Z_0(s)=-\int_s^\infty K(s,s')N(Z_0(s'))\,\mathrm{d}s'$ (all boundary terms vanish). With $m(s)=\sup_{[s,\infty)}|Z_0|$ this yields $m(s)\le Cm(s)\,e^{-\nu s}\int_s^\infty e^{\nu s'}m(s')\,\mathrm{d}s'$, and the last factor tends to 0, so $Z_0\equiv 0$ near $+\infty$ and hence everywhere, contradicting $Y_0(s)\to\infty$ as $s\to-\infty$. Finally write $(C_1,C_2)=A_1(\cos A_2,-\sin A_2)$. □

4.2. **The inner region of v_c.** Set $V_c(r):=c^{-1}v_c(c^{3/2}r)$; then $Y_c(s)=e^{-2(s-\sigma)/3}V_c(e^{s-\sigma})$ and, by [7, (2.9)],

$$V_c''+\frac{n-1}{r}V_c'-\frac{1}{V_c^2}=c^3\Big(r^2V_c''+\frac{2}{3}rV_c'-\frac{2}{9}V_c\Big),\qquad V_c(0)=1,\ V_c'(0)=0. \tag{28}$$

Lemma 4.2 (inner stability)**.** *For every $S_0\ge 1$ there are $C(S_0)$ and $c_1(S_0)>0$ such that for $0<c\le c_1(S_0)$*

$$\sup_{0\le r\le e^{S_0}}\big(|V_c-V_0|+|V_c'-V_0'|\big)\le C(S_0)\,c^3,$$
$$\sup_{\sigma-1\le s\le\sigma+S_0}\big(|Y_c(s)-Y_0(s-\sigma)|+|Y_c'(s)-Y_0'(s-\sigma)|\big)\le C(S_0)\,c^3.$$

In particular $v_c>0$ and $v_c'>0$ on $(0,c^{3/2}e^{S_0}]$, and $v_c(\eta)=cV_0(\eta/c^{3/2})(1+O(c^3))$ there.

Proof. Solving (28) for V_c'', $V_c''=\big[-\frac{n-1}{r}V_c'+V_c^{-2}+c^3(\frac{2}{3}rV_c'-\frac{2}{9}V_c)\big]/(1-c^3r^2)$, which on $[0,e^{S_0}]$ is a regular (in c^3) perturbation of the equation for V_0 once $c^3e^{2S_0}\le\frac12$; the point $r=0$ is handled through the integral form $V(r)=1+\int_0^r\zeta^{1-n}\int_0^\zeta\xi^{n-1}F(\xi,V,V')\,\mathrm{d}\xi\,\mathrm{d}\zeta$ exactly as in [7, Lemma A.3]. Continuous dependence on parameters (Gronwall on a compact interval) gives the first bound; the second follows since $r=e^{s-\sigma}\in[e^{-1},e^{S_0}]$ is bounded away from 0 and ∞, and the last statement from $V_0\ge 1$, $V_0'>0$ on $(0,\infty)$ together with $V_c'(r)=\frac{r}{n}(1+O(r^2))>0$ for small r. □

4.3. **The singular coefficient near the singular profile.** We normalise the singular coefficient of solutions of the linearised equation as in Proposition 2.4: for a solution Z of $\mathcal{L}Z=0$ on $(-\delta,0)$, $\beta_{\mathrm{lin}}[Z]$ is the number B in $Z=A\chi_{\mathrm{reg}}+B\chi_{\mathrm{sing}}$; since $1-\eta=|s|(1+O(s))$ and $\eta^{2/3}=1+O(s)$, this is the coefficient of $(1-\eta)^\alpha$ in $\eta^{2/3}Z$, so $\beta_{\mathrm{lin}}[\Psi_j]=\sigma_j$. Fix $\delta_1\in(0,1)$ and $x_0:=1-e^{-\delta_1}$.

Lemma 4.3 (linearisation of the shooting functional)**.** *There are $r_0,C>0$ and a linear form $\ell:\mathbb{R}^2\to\mathbb{R}$ with the following property. If $Y=a+Z$ solves (5) on $[-\delta_1,0)$ and*

$d := |Z(-\delta_1)| + |Z'(-\delta_1)| \le r_0$, then $v = \eta^{2/3} Y$ satisfies the hypotheses of Proposition 2.4, and its singular coefficient obeys

$$\big|\beta[v] - \ell\big(Z(-\delta_1), Z'(-\delta_1)\big)\big| \le C d^2, \qquad \ell\big(\Psi_j(-\delta_1), \Psi_j'(-\delta_1)\big) = \sigma_j \quad (j = 1, 2).$$

Proof. By Lemma 2.6 and its proof, the map $\Pi : (c_1, \beta') \mapsto (v, \psi)(x_0; c_1, \beta')$ is an analytic diffeomorphism from a neighbourhood of $(a, 0)$ onto a neighbourhood of the data of v_{s} at x_0, and $\beta[v] = -\frac{2^{\alpha-1}}{\alpha} \operatorname{pr}_2 \Pi^{-1}\big((v, \psi)(x_0)\big)$. The data $(v, \psi)(x_0)$ depend linearly on $(Z, Z')(-\delta_1)$ (through $v = \eta^{2/3}(a + Z)$ and (8), (13)), so $\mathfrak{B} := \beta \circ (\text{data map})$ is analytic near $(0, 0)$ with $\mathfrak{B}(0, 0) = \beta[v_{\mathrm{s}}] = 0$, and $\ell := D\mathfrak{B}(0, 0)$ gives the first estimate by Taylor's theorem. For the second, differentiate the one-parameter family $v^\varepsilon :=$ Π-solution with parameters $(a, 0) + \varepsilon\, D\Pi(a, 0)^{-1}(\text{data of } \eta^{2/3}\Psi_j)$: its ε-derivative at $\varepsilon = 0$ is a solution of the linearisation of (14) at v_{s} with the data of Ψ_j, hence equals Ψ_j (in flux variables), and the ε-derivative of its β'-parameter is by construction the coefficient of $x^{\alpha-1}$ in its ψ-component, i.e. $-\frac{\alpha}{2^{\alpha-1}}\sigma_j$. Thus $\ell(\text{data of } \Psi_j) = \sigma_j$. □

4.4. **Matching: asymptotics of $\beta(c)$.**

Proposition 4.4 (matching)**.** *Let $2 \le n \le 7$ and $\mathcal{A}(c) := A_1 c^{3\nu/2} = A_1 e^{\nu\sigma}$. There is $C > 0$ such that for every $S_0 \ge 1$ there is $c_1(S_0) > 0$ with*

(29)
$$\beta(c) = -\rho_0\, \mathcal{A}(c)\Big[\sin\Big(\frac{3b}{2}\ln\frac{1}{c} - \vartheta\Big) + E(c, S_0)\Big], \qquad |E(c, S_0)| \le C\big(e^{-\nu S_0} + c^3 e^{(2+\nu)S_0} + c^{3\nu/2}\big),$$

for all $0 < c \le c_1(S_0)$, with $\vartheta = B_2 - A_2$ as in (21). Moreover, for such c and with $s_m := \sigma + S_0$,

(30)
$$\begin{aligned} Z_c(s) &= c_1\Psi_1(s) + c_2\Psi_2(s) + Z_N(s), \qquad \theta := A_2 - b\sigma, \\ (c_1, c_2) &= \mathcal{A}(c)\big[(\cos\theta, -\sin\theta) + O(e^{-\nu S_0} + c^3 e^{(2+\nu)S_0})\big], \end{aligned}$$

where $|Z_N(s)| + |Z_N'(s)| \le C e^{-\nu S_0}\mathcal{A}(c) e^{-\nu s}$ on $[s_m, -\delta_1]$, and $|Z_c| + |Z_c'| \le C\mathcal{A}(c)$ on $[-\delta_1, 0)$.

Proof. Step 1: data at s_m. By Lemma 4.2 and (27),

$$\begin{pmatrix} Z_c \\ Z_c' \end{pmatrix}(s_m) = A_1 e^{-\nu S_0}\Big[\begin{pmatrix} \cos \\ -\nu\cos - b\sin \end{pmatrix}(bS_0 + A_2) + O\big(e^{-\nu S_0} + c^3 e^{\nu S_0}\big)\Big].$$

Step 2: the linear part. Let $W(s)$ be the matrix with columns $(\Psi_j, \Psi_j')(s)$, so that by (19) $W(s) = e^{-\nu s}[W_0(s) + O(e^{2s})]$ with $\det W_0 \equiv b$. Define $(c_1, c_2)^T := W(s_m)^{-1}(Z_c, Z_c')(s_m)$. Since $A_1 e^{-\nu(s-\sigma)}\cos(b(s-\sigma)+A_2) = \mathcal{A}(c)e^{-\nu s}[\cos\theta\cos bs - \sin\theta\sin bs]$, the inner tail has data $\mathcal{A}(c)e^{-\nu s_m} W_0(s_m)(\cos\theta, -\sin\theta)^T$ at s_m, and Step 1 gives (30) (the term $O(e^{2s_m}) = O(c^3 e^{2S_0})$ from the Levinson error is absorbed).

Step 3: the nonlinear part on $[s_m, -\delta_1]$. Z_N solves $\mathcal{L}Z_N = -N(Z_c)$ with zero data at s_m, hence $Z_N(s) = \int_{s_m}^s \mathcal{K}(s, s')\,\frac{-N(Z_c(s'))}{1 - e^{2s'}}\,\mathrm{d}s'$ with $\mathcal{K}(s, s') = [\Psi_1(s')\Psi_2(s) - \Psi_2(s')\Psi_1(s)]/\mathcal{W}(s')$, $\mathcal{W} = \Psi_1\Psi_2' - \Psi_2\Psi_1'$. As $\mathcal{W}' = -\frac{n - \frac{2}{3} - e^{2s}}{1 - e^{2s}}\mathcal{W}$, we have $\mathcal{W}(s) \asymp e^{-2\nu s}$ on $(-\infty, -\delta_1]$, and therefore $|\mathcal{K}(s, s')| + |\partial_s\mathcal{K}(s, s')| \le C e^{-\nu(s-s')}$ for $s_m \le s' \le s \le -\delta_1$. Assume (bootstrap) $|Z_N| \le \mathcal{A}(c)e^{-\nu s}$ on $[s_m, s^*]$. Then $|Z_c| \le C\mathcal{A}(c)e^{-\nu s} \le a/2$ there (as $\mathcal{A}(c)e^{-\nu s_m} = A_1 e^{-\nu S_0}$ is small), so $|N(Z_c(s'))| \le C\mathcal{A}(c)^2 e^{-2\nu s'}$ and

$$|Z_N(s)| + |Z_N'(s)| \le C\mathcal{A}(c)^2 \int_{s_m}^s e^{-\nu(s-s')} e^{-2\nu s'}\,\mathrm{d}s' \le \frac{C}{\nu}\mathcal{A}(c)^2 e^{-\nu s} e^{-\nu s_m} = \frac{CA_1}{\nu} e^{-\nu S_0}\,\mathcal{A}(c) e^{-\nu s},$$

which improves the bootstrap assumption once $e^{-\nu S_0}$ is small, and proves the bound on Z_N on all of $[s_m, -\delta_1]$.

Step 4: the cone. By Step 3, $d := |Z_c(-\delta_1)| + |Z_c'(-\delta_1)| \le C\mathcal{A}(c)$, so Lemma 4.3 applies (recall $v_c = \eta^{2/3} Y_c$ reaches the cone by Lemma 2.1) and gives, with ℓ linear,

$$\beta(c) = \ell\big((Z_c, Z_c')(-\delta_1)\big) + O(\mathcal{A}(c)^2) = c_1\sigma_1 + c_2\sigma_2 + O\big(e^{-\nu S_0}\mathcal{A}(c)\big) + O(\mathcal{A}(c)^2).$$

Inserting (30) and $(\sigma_1,\sigma_2)=\rho_0(\cos\varphi,\sin\varphi)$, $c_1\sigma_1+c_2\sigma_2=\rho_0\mathcal{A}(c)[\cos(\theta+\varphi)+O(e^{-\nu S_0}+c^3e^{(2+\nu)S_0})]$, and $\theta+\varphi=A_2-b\sigma+\frac{\pi}{2}-B_2=\frac{3b}{2}\ln\frac{1}{c}-\vartheta+\frac{\pi}{2}$, so $\cos(\theta+\varphi)=-\sin(\frac{3b}{2}\ln\frac{1}{c}-\vartheta)$. The bound on Z_c on $[-\delta_1,0)$ follows from the contraction estimates of [7, Lemma A.2] (continuous dependence up to the cone), or from the analytic parametrisation Π of Lemma 4.3. □

Remark 4.5. The whole construction depends analytically on σ, and each estimate above may be differentiated with respect to σ (the derivative of the inner tail $\partial_\sigma[Y_0(s-\sigma)]=-Y_0'(s-\sigma)$ has the same form (27), the Levinson errors are uniform, and the bootstrap of Step 3 closes equally for $\partial_\sigma Z_N$). This yields (29) in C^1 with respect to $\ln c$, i.e. $\frac{d}{d\ln c}\big[\beta(c)/(\rho_0\mathcal{A}(c))\big]=\frac{3b}{2}\cos(\frac{3b}{2}\ln\frac{1}{c}-\vartheta)+o(1)$, and hence that β has exactly one zero in each window J_k, that zero being transversal. We do not carry this out here.

5. Proof of Theorem 1.2

Choice of parameters. Let C be the constant of Proposition 4.4. Given $\varepsilon\in(0,\frac{1}{10})$ choose $S_0(\varepsilon)$ with $Ce^{-\nu S_0}\le\varepsilon/3$ and then $c(\varepsilon)\le c_1(S_0(\varepsilon))$ so small that $C(c^3e^{(2+\nu)S_0}+c^{3\nu/2})\le\varepsilon/3$ for $c\le c(\varepsilon)$; thus $|E|\le\varepsilon$ in (29) for $0<c\le c(\varepsilon)$. Let $\Theta(c):=\frac{3b}{2}\ln\frac{1}{c}-\vartheta$, an increasing function of $\ln\frac{1}{c}$, and $c_0:=c(\frac{1}{10})$.

(i) Zeros of β. Fix $\varepsilon_k\downarrow 0$ and choose k_0 and an increasing sequence of indices such that $c\le c(\varepsilon_k)$ whenever $\Theta(c)\ge\pi k-\frac{\pi}{2}$, $k\ge k_0$ (possible since $\Theta(c)\to\infty$ as $c\to 0$). Put $\varpi_k:=\arcsin(2\varepsilon_k)$ and $J_k:=\{c:|\Theta(c)-\pi k|\le\varpi_k\}$. If $c\in(0,c_0]$ lies in no J_k, then $|\sin\Theta(c)|>2\varepsilon_k\ge 2|E|$ for the relevant k, so $\beta(c)\neq 0$. At the two points $c_k^\pm$ with $\Theta(c_k^\pm)=\pi k\pm\frac{\pi}{2}$ one has $\sin\Theta=\pm1$ and $|E|\le\frac{1}{10}$, so $\beta(c_k^+)\beta(c_k^-)<0$; by continuity of β (Lemma 2.6) there is a zero in (c_k^+,c_k^-), and by the previous sentence it lies in J_k. Since β is analytic and changes sign, its zeros in $(0,c_0]$ are isolated; hence $\mathcal{C}$ is discrete, $\mathcal{C}\subset\bigcup_{k\ge k_0}J_k$, $\mathcal{C}\cap J_k\neq\emptyset$, and (11) follows from $\Theta(c_k)=\pi k+O(\varpi_k)$. By Definition 2.5, $c\in\mathcal{C}$ exactly when v_c extends to a C^∞ solution across the cone; that the extension is global and smooth is shown next.

(ii) Positivity, monotonicity and the crossing of c^*. Let $c\in\mathcal{C}\cap(0,c_0]$ and $S_0=S_0(\frac{1}{10})$. On $[0,c^{3/2}e^{S_0}]$, Lemma 4.2 gives $v_c>0$, $v_c'>0$ and $v_c\le c(1+e^{2S_0}/2n)(1+O(c^3))<c^*$. On $[c^{3/2}e^{S_0},1)$, Proposition 4.4 gives $|Z_c|\le C\mathcal{A}(c)$, i.e. $|v_c-v_{\rm s}|\le C\mathcal{A}(c)\eta^{2/3}$, so $v_c\le a+C\mathcal{A}(c)<c^*$ for c small (Lemma 1.1). Since $v_c'(\eta)=\eta(c^{-2}-\frac{2}{9}c)/n+O(\eta^3)>0$ near 0 and $v_c<c^*$ on $[0,1)$, Lemma 3.1 (no local maximum below c^* inside the cone) shows $v_c'>0$ on $(0,1)$. At the cone, $\beta(c)=0$ and v_c coincides near $\eta=1$ with the analytic branch $v_{\rm reg}(\cdot;v_c(1))$, where $v_c(1)=a+O(\mathcal{A}(c))$. By Lemma 2.3 and regular continuous dependence on $[1+\delta,\eta_0]$, $\eta_0:=\eta^*+1$, $v_{\rm reg}(\cdot;v_c(1))\to v_{\rm s}$ in $C^1([1,\eta_0])$ as $c\to 0$. As $v_{\rm s}$ crosses c^* transversally at $\eta^*\in(1,\eta_0)$ (Lemma 1.1: $v_{\rm s}'(\eta^*)>0$), for c small v_c crosses c^* at some $\eta_c^{**}\to\eta^*$ with $v_c<c^*$ on $[1,\eta_c^{**})$. Proposition 3.2 now applies: v_c is a global solution, $v_c'>0$ on $(0,\infty)$, $v_c>c^*$ beyond η_c^{**}. Analyticity on $(0,\infty)$ holds since (1) is analytic and non-degenerate away from $\eta=1$, and across $\eta=1$ by Lemma 2.3; smoothness at $\eta=0$ is [7, Lemma A.3]. This proves (ii).

(iii) is Proposition 3.2(iii). For $b_k\to a$, note that (Y_c,∂_tY_c) at $t=1/\eta_0$ converges to $(a,0)$ by the C^1 convergence on $[1,\eta_0]$, and that $Y\equiv a$ is the solution of the regular equation (26) with these limiting data on $[0,1/\eta_0]$; continuous dependence gives $Y_c(0)=b\to a$.

(iv) is Lemma 4.2 (with $V_0\ge 1$) and the bound $|Z_c|\le C\mathcal{A}(c)$ on $[\sigma+S_0,0]$ obtained in the proof of Proposition 4.4, together with (27) on $[\sigma+S_0,\sigma+S_0']$ for any $S_0'>S_0$ (where $|Z_c-A_1e^{-\nu(s-\sigma)}\cos(b(s-\sigma)+A_2)|\le C(S_0')(c^3+e^{-2\nu S_0})$) and the continuity of $v_{\rm reg}(\cdot;c_1)$ in c_1 up to $\eta=1+\delta_0$.

(v) Intersection count. The zeros of $v_c-v_{\rm s}$ in $(0,1)$ are the zeros of Z_c in $(-\infty,0)$. Choose S_0 with $Y_0(S_0)\neq a$ and $bS_0+A_2\notin\frac{\pi}{2}+\pi\mathbb{Z}$, and $M>\delta_1$ with $\Psi_{\rm reg}(-M)\neq 0$ and $-bM+B_2\notin\frac{\pi}{2}+\pi\mathbb{Z}$. On $(-\infty,\sigma+S_0]$ the zeros of Z_c are, by Lemma 4.2, in one-to-one correspondence with the zeros of Y_0-a in $(-\infty,S_0]$ (finitely many, all simple because

$Y_0' = 0 = Y_0 - a$ would force $Y_0 \equiv a$), a number independent of c. On $[\sigma + S_0, -M]$, by (30) and (19), $Z_c = \mathcal{A}(c)e^{-\nu s}[\cos(b(s-\sigma) + A_2) + O(e^{-\nu S_0} + e^{-2M})]$ in C^1, so its zeros correspond to the solutions of $b(s-\sigma) + A_2 \in \frac{\pi}{2} + \pi\mathbb{Z}$ in that interval, whose number is $\lfloor (\pi k + \pi/2 - bM - bS_0 + B_2 - A_2 + \pi/2)/\pi \rfloor$ up to a fixed shift when $c \in J_k$ with k large (use $-b\sigma = \vartheta + \pi k + O(\varpi_k)$), and it increases by exactly one from J_k to J_{k+1} because the endpoints stay at a fixed distance from the zeros. On $[-M, 0)$, $Z_c/\mathcal{A}(c)$ is C^1-close to $\pm\rho_0^{-1}\Psi_{\mathrm{reg}}$ for $c \in J_k$, k large (the singular component is $O(\varpi_k)$), so the number of zeros is that of Ψ_{reg} on $[-M, 0)$, independent of k and of the sign. Summing the three counts proves (v). □

6. Numerical confirmation

The structure of the proof suggests a robust *two-point shooting* scheme, which we used to check Theorem 1.2 (scripts in the accompanying `numerics/` directory). For a given c the regular solution v_c is integrated from $\eta = 0$ (started from its Taylor expansion $v = c + \frac{c^{-2} - \frac{2}{9}c}{2n}\eta^2 + \dots$) to a matching point $\eta_m = 0.85$, in the variable $s = \ln\eta$ with a high-order Runge–Kutta method. Independently, for a given value $c_1 = v(1)$ the analytic branch $v_{\mathrm{reg}}(\cdot; c_1)$ of Lemma 2.3 is generated by its Taylor series at the cone (the recursion of that proof) and integrated backwards from $\eta = 1 - \delta$ to η_m. The function

$$g(c) = v_c'(\eta_m) - v_{\mathrm{reg}}'(\eta_m; c_1(c)), \qquad c_1(c) \text{ defined by } v_{\mathrm{reg}}(\eta_m; c_1(c)) = v_c(\eta_m),$$

vanishes exactly at the smooth global profiles and is a numerical proxy for $\beta(c)$; its zeros are located by bisection after a scan in $\ln c$. Each profile is then continued beyond the cone from the Taylor series at $\eta = 1$ up to $\eta = 10^4$ to read off b_k and to check monotonicity. The constants A_1, A_2 (least-squares fit of $Y_0 - a$ on a window in s) and B_2 (Levinson asymptotics of the numerically integrated Ψ_{reg}) are computed separately, and the last two columns of Table 1 compare the zeros with the leading-order prediction $c_k^{\mathrm{formal}} = \exp\big(-\frac{2}{3b}(\vartheta + \pi k)\big)$ of (29), i.e. with (2.33) of [7] with πk in place of $2\pi k$.

The agreement between the last two columns improves like a power of c_k, as predicted by (29); for $n = 3$, $k = 5$ the leading-order matching already gives seven correct digits. The phases used are $(A_2, B_2) = (2.022, 0.951)$, $(0.860, -0.426)$, $(-0.152, 1.426)$, $(-1.021, 0.144)$, $(-1.715, 1.956)$, $(-2.136, 0.470)$ for $n = 2, \dots, 7$.

Acknowledgement

This note was drafted with the help of an AI assistant (Claude); all statements and proofs were checked by the author.

References

[1] P. Bizoń, D. Maison, A. Wasserman, Self-similar solutions of semilinear wave equations with a focusing nonlinearity, *Nonlinearity* 20 (2007) 2061–2074.
[2] C. Budd, Y.-W. Qi, The existence of bounded solutions of a semilinear elliptic equation, *J. Differential Equations* 82 (1989) 207–218.
[3] E. A. Coddington, N. Levinson, *Theory of Ordinary Differential Equations*, McGraw–Hill, 1955.
[4] C. Collot, P. Raphaël, J. Szeftel, On the stability of type I blow up for the energy super critical heat equation, *Mem. Amer. Math. Soc.* 260 (2019), no. 1255.
[5] W. Dai, T. Duyckaerts, Self-similar solutions of focusing semi-linear wave equations in $\mathbb{R}^N$, *J. Evol. Equ.* 21 (2021) 4703–4740.
[6] M. S. P. Eastham, *The Asymptotic Solution of Linear Differential Systems*, Oxford Univ. Press, 1989.
[7] H. Gimperlein, R. He, A. A. Lacey, Quenching for a semi-linear wave equation for micro-electro-mechanical systems, *Proc. R. Soc. A* 478 (2022) 20220490.
[8] E. Hille, *Ordinary Differential Equations in the Complex Domain*, Wiley, 1976.
[9] K. Iwasaki, H. Kimura, S. Shimomura, M. Yoshida, *From Gauss to Painlevé*, Vieweg, 1991.
[10] N. I. Kavallaris, A. A. Lacey, C. V. Nikolopoulos, D. E. Tzanetis, On the quenching behaviour of a semilinear wave equation modelling MEMS technology, *Discrete Contin. Dyn. Syst.* 35 (2015) 1009–1037.

n	k	c_k	$v_k(1)$	b_k	zeros of $v_k - v_{\mathrm{s}}$	c_k^{formal}	rel. err.
2	1	0.23969913	1.271208	1.498995	1	0.23135200	$3.5 \cdot 10^{-2}$
2	2	0.02499324	1.314581	1.291049	2	0.02509074	$-3.9 \cdot 10^{-3}$
2	3	0.00272231	1.309914	1.312479	3	0.00272116	$4.2 \cdot 10^{-4}$
3	1	0.40504140	0.940347	1.154685	1	0.41437736	$-2.3 \cdot 10^{-2}$
3	2	0.09335645	0.967509	0.951396	2	0.09326298	$1.0 \cdot 10^{-3}$
3	3	0.02098872	0.965342	0.966526	3	0.02099049	$-8.4 \cdot 10^{-5}$
3	4	0.00472431	0.965500	0.965413	4	0.00472428	$6.1 \cdot 10^{-6}$
3	5	0.00106328	0.965489	0.965495	5	0.00106328	$-3.9 \cdot 10^{-7}$
4	0	0.47418175	0.837698	1.006057	0	0.51782309	$-9.2 \cdot 10^{-2}$
4	1	0.13977134	0.824947	0.816470	1	0.13970024	$5.1 \cdot 10^{-4}$
4	2	0.03768561	0.825502	0.825820	2	0.03768885	$-8.6 \cdot 10^{-5}$
4	3	0.01016786	0.825481	0.825469	3	0.01016784	$2.0 \cdot 10^{-6}$
4	4	0.00274312	0.825482	0.825482	4	0.00274312	$-4.4 \cdot 10^{-8}$
4	5	0.00074005	0.825482	0.825482	5	0.00074005	$4.4 \cdot 10^{-7}$
5	0	0.51205398	0.787565	0.914301	0	0.61981854	$-2.1 \cdot 10^{-1}$
5	1	0.16949413	0.741674	0.736655	1	0.17074041	$-7.4 \cdot 10^{-3}$
5	2	0.04702869	0.742357	0.742433	2	0.04703358	$-1.0 \cdot 10^{-4}$
5	3	0.01295623	0.742346	0.742345	3	0.01295627	$-3.0 \cdot 10^{-6}$
5	4	0.00356905	0.742346	0.742346	4	0.00356904	$2.7 \cdot 10^{-6}$
6	0	0.18793550	0.685459	0.681545	0	0.19367157	$-3.1 \cdot 10^{-2}$
6	1	0.04750634	0.685005	0.685020	1	0.04752238	$-3.4 \cdot 10^{-4}$
6	2	0.01166082	0.685007	0.685007	2	0.01166086	$-3.3 \cdot 10^{-6}$
6	3	0.00286258	0.685007	0.685007	3	0.00286130	$4.5 \cdot 10^{-4}$
7	0	0.19720292	0.643996	0.640182	0	0.21866791	$-1.1 \cdot 10^{-1}$
7	1	0.03495458	0.642078	0.642079	1	0.03497245	$-5.1 \cdot 10^{-4}$
7	2	0.00560440	0.642078	0.642078	2	0.00559329	$2.0 \cdot 10^{-3}$

TABLE 1. Smooth global profiles found by two-point shooting. k is the index of the window J_k of Theorem 1.2 (the nearest integer to $\Theta(c_k)/\pi$); in every case it coincides with the number of zeros of $v_k - v_{\mathrm{s}}$ in $(0,1)$. The trivial profile $c = c^*$ and, for $n \geq 4$, one further large-c profile outside the asymptotic regime ($k < 0$) are omitted, as are zeros with $c^{3\nu/2} < 10^{-11}$, where the deviation from v_{s} is below double-precision resolution. All profiles are strictly increasing and $b_k \to a$ ($a = 1.3104, 0.9655, 0.8255, 0.7423, 0.6850, 0.6421$ for $n = 2, \dots, 7$). The ratios c_{k+1}/c_k converge to $e^{-2\pi/(3b)} = 0.1085, 0.2251, 0.2698, 0.2755, 0.2454, 0.1599$.

[11] L. A. Lepin, Self-similar solutions of a semilinear heat equation, *Mat. Model.* 2 (1990) 63–74.

[12] F. Merle, P. Raphaël, I. Rodnianski, J. Szeftel, On the implosion of a compressible fluid I: Smooth self-similar inviscid profiles, *Ann. of Math.* 196 (2022) 567–778.

[13] F. Merle, P. Raphaël, I. Rodnianski, J. Szeftel, On blow up for the energy super critical defocusing nonlinear Schrödinger equations, *Invent. Math.* 227 (2022) 247–413.

[14] F. Merle, H. Zaag, Determination of the blow-up rate for the semilinear wave equation, *Amer. J. Math.* 125 (2003) 1147–1164.

[15] F. Merle, H. Zaag, Determination of the blow-up rate for a critical semilinear wave equation, *Math. Ann.* 331 (2005) 395–416.

[16] W. C. Troy, The existence of bounded solutions of a semilinear heat equation, *SIAM J. Math. Anal.* 18 (1987) 332–336.

INSTITUTO DE CIENCIAS MATEMÁTICAS (ICMAT), MADRID